# Quizrr Chapter-Wise Test for JEE Advanced - 2025

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**Q.1**

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, $\sum_{r=0}^{n}\left((r+1)^2\right) C_r = 2^{n-2}f(n)$ and if the roots of the equation $f(x) = 0$ are $\alpha$ and $\beta$, then the value of $\alpha^2 + \beta^2$ is equal to (where $C_r$ denotes ${}^nC_r$)

**(A)** $13$

**(B)** $10$

**(C)** $17$

**(D)** $20$

**Q.2**

If $x^m$ occurs in the expansion of $\left(x + \frac{1}{x^2}\right)^{2n}$, the coefficient of $x^m$ is -

**(A)** $\frac{(2n)!}{m!(2n-m)!}$

**(B)** $\frac{(2n)!3!3!}{(2n-m)!}$

**(C)** $\frac{(2n)!}{\left(\frac{2n-m}{3}\right)!\left(\frac{4n+m}{3}\right)!}$

**(D)** None of these

**Q.3**

If the third term in the expansion of $\left[x + x^{\log_{10} x}\right]^5$ is equal to $10,00,000$ then $x$ equals-

**(A)** $10$

**(B)** $10^2$

**(C)** $10^3$

**(D)** No such $x$ exists

**Q.4**

The greatest integer in the expansion of $(1+x)^{2n+2}$ is -

**(A)** $\frac{(2n)!}{(n!)^2}$

**(B)** $\frac{(2n+2)!}{[(n+1)!]^2}$

**(C)** $\frac{(2n+2)!}{n!(n+1)!}$

**(D)** $\frac{(2n)!}{n!(n+1)!}$

**Q.5**

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The greatest integer which divides $101^{100}-1$ is -

**(A)** 100

**(B)** 1000

**(C)** 10,000

**(D)** 100,000

**Q.6**

The sum of the rational terms in the expansion of $\left(\sqrt{2}+3^{1/5}\right)^{10}$ is equal to

**(A)** 40

**(B)** 41

**(C)** 42

**(D)** 0

**Q.7**

If the coefficient of $x^{10}$ in the expansion of $\left(1+x^2-x^3\right)^8$ is $68\lambda$ then $\lambda$ equals

**(A)** 7

**(B)** 5

**(C)** 9

**(D)** 4

**Q.8**

If $C_0,\ C_1,\ C_2,\ \ldots\ldots,\ C_{20}$ are the binomial coefficients in the expansion of $(1+x)^{20}$, then the value of $\frac{C_1}{C_0}+2\frac{C_2}{C_1}+3\frac{C_3}{C_2}+\ldots\ldots+19\frac{C_{19}}{C_{18}}+20\frac{C_{20}}{C_{19}}$ is equal to (where $C_r$ represents ${}^nC_r$)

**(A)** 120

**(B)** 210

**(C)** 180

**(D)** 240

**Q.9**

If $(4+\sqrt{15})^{\mathrm{n}}=\mathrm{I}+\mathrm{f}$, where $\mathrm{n}$ is an odd natural number, I is an integer and $0<\mathrm{f}<1$, then $-$

**(A)** I is a natural number

**(B)** I is an even integer

**(C)** $(\mathrm{I}+\mathrm{f})(1-\mathrm{f})=1$

**(D)** $1-\mathrm{f}=(4-\sqrt{15})^{\mathrm{n}}$

**Q.10**

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If the coefficients of $r^{\text{th}}$ , $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms in the expansion of $(1+x)^{14}$ are in A.P., then $r$ is/are -

**(A)** 5

**(B)** 12

**(C)** 10

**(D)** 9

**Q.11**

The expansion of $(1+x)^n$ has 3 consecutive terms with coefficients in the ratio $1:2:3$ and can be written in the form ${}^nC_k : {}^nC_{k+1} : {}^nC_{k+2}$. Values greater than the sum of all possible values of $(n+k)$ is -

**(A)** 18

**(B)** 21

**(C)** 28

**(D)** 32

**Q.12**

**Passage:**
An equation $a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = 0$ has roots ${}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \ldots\ldots, {}^{99}C_{99}$.
**Question:**
The value of $a_{99}$ is equal to -

**(A)** $2^{98}$

**(B)** $2^{99}$

**(C)** $-2^{99}$

**(D)** None of these

**Q.13**

**Passage:**
An equation $a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = 0$ has roots ${}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \ldots\ldots, {}^{99}C_{99}$.
**Question:**
The value of $a_{98}$ is -

**(A)** $\frac{2^{198} - {}^{198}C_{99}}{2}$

**(B)** $\frac{2^{198} + {}^{198}C_{99}}{2}$

**(C)** $2^{99} - {}^{99}C_{49}$

**(D)** None of these

**Q.14**

**Passage:**
An equation $a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = 0$ has roots ${}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \ldots\ldots, {}^{99}C_{99}$.
**Question:**
The value of $\left({}^{99}C_0\right)^2 + \left({}^{99}C_1\right)^2 + \ldots. + \left({}^{99}C_{99}\right)^2$ is equal to -

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**(A)** $2a_{98} - a_{99}^2$

**(B)** $a_{99}^2 - a_{98}$

**(C)** $a_{99}^2 - 2a_{98}$

**(D)** None of these

**Q.15**

**Passage:**
The $2^{nd}$ , $3^{rd}$ and $4^{th}$ terms in the expansion of $(x + a)^n$ are 240,720 and 1080, respectively —
**Question:**
The value of $(x - a)^n$ can be —

**(A)** 64

**(B)** -1

**(C)** -32

**(D)** None of these

**Q.16**

**Passage:**
The $2^{nd}$ , $3^{rd}$ and $4^{th}$ terms in the expansion of $(x + a)^n$ are 240,720 and 1080, respectively —
**Question:**
The value of least term in the expansion is -

**(A)** 16

**(B)** 160

**(C)** 32

**(D)** 81

**Q.17**

**Passage:**
The $2^{nd}$ , $3^{rd}$ and $4^{th}$ terms in the expansion of $(x + a)^n$ are 240,720 and 1080, respectively —
**Question:**
The sum of odd-numbered terms is -

**(A)** 1664

**(B)** 2376

**(C)** 1562

**(D)** 1486

**Q.18**

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Match the column -

| Column I | Column II |
|---|---|
| (A) $^{32}C_0^2 - {}^{32}C_1^2 + {}^{32}C_2^2 - ..... + {}^{32}C_{32}^2 =$ | (p) $^{63}C_{32}$ |
| (B) $^{32}C_0^2 + {}^{32}C_1^2 + {}^{32}C_2^2 - ..... + {}^{32}C_{32}^2 =$ | (q) $^{32}C_{16}$ |
| (C) $\frac{1}{32}(1 \times {}^{32}C_1^2 + 2 \times {}^{32}C_2^2 + ..... + 32 \times {}^{32}C_{32}) =$ | (r) 0 |
| (D) $^{31}C_0^2 - {}^{31}C_1^2 + {}^{31}C_2^2 - ..... - {}^{31}C_{31}^2 =$ | (s) $^{64}C_{32}$ |

**(A)** (A) - p, (B) - r, (C) - s, (D) - p

**(B)** (A) - q, (B) - s, (C) - p, (D) - r

**(C)** (A) - s, (B) - p, (C) - r, (D) - q

**(D)** (A) - p, (B) - q, (C) - r, (D) - s

**Q.19**

In the binomial expansion of $\left(1-\frac{1}{x}+3x^5\right)\left(2x^2-\frac{1}{x}\right)^8$, the term independent of $x$ is

**Q.20**

$\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5$ is a polynomial of the order of -

## Answers & Solutions

**Q.1 Answer:**

17

**Solution:**

$(1+x)^n = \sum_{r=0}^{n} C_r x^r$

$\Rightarrow x(1+x)^n = \sum_{r=0}^{n} C_r x^{r+1}$

**Differentiating w.r.t. $x$ we get**

$xn(1+x)^{n-1} + (1+x)^n = \sum_{r=0}^{n}(r+1)C_r x^r$

**Again multiplying both sides by $x$**

$(1+x)^{n-1}(nx+1+x)x = \sum_{r=0}^{n}(r+1)C_r x^{r+1}$

**Again differentiating w.r.t. $x$ we get,**

$\frac{d}{dx}\left((1+x)^{n-1}\left(nx^2+x^2+x\right)\right) = \sum_{r=0}^{n}\left((r+1)^2\right)C_r x^r$

$\sum_{r=0}^{n}\left((r+1)^2\right)C_r x^r = (1+x)^{n-1}(2nx+2x+1) + \left(nx^2+x^2+x\right)(n-1)(1+x)^{n-2}$

**Putting $x=1$ on both sides, we get,**

$\sum_{r=0}^{n}\left((r+1)^2\right)C_r = 2^{n-1}(2n+3) + (n+2)\cdot(n-1)2^{n-2} = 2^{n-2}\left(n^2+5n+4\right)$

**Now,** $f(x) = x^2+5x+4 = (x+1)(x+4) \Rightarrow \alpha = -1,\ \beta = -4$

**Hence,** $\alpha^2+\beta^2 = 17$

**Q.2 Answer:**

$\frac{(2n)!}{\left(\frac{2n-m}{3}\right)!\left(\frac{4n+m}{3}\right)!}$

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**Solution:**

**The general term in the expansion of the given expression is**

$\left(\ 1\ \right)^{r}$

**For the coefficient of $x^m$, we must have $2n-3r=m \Rightarrow r=\frac{2n-m}{3}$**
**So, coefficient of $x^m = {}^{2n}C_{\frac{2n-m}{3}} = \frac{(2n)!}{\left(\frac{2n-m}{3}\right)!\left(\frac{4n+m}{3}\right)!}$**

**Q.3 Answer:**

10

**Solution:**

**Here $T_3 = {}^5C_2 x^3 \left(x \log_{10}^x\right)^2 = 10^6$ or $x^3 x^{2\log_{10}^x} = 10^5$**
**Taking log of both sides, we get $3\log_{10} x + 2\left(\log_{10} x\right)^2 = 5$**
**or $2\left(\log_{10} x\right)^2 + 5\log_{10} x - 2\log_{10} x - 5 = 0$**
**or $\left(\log_{10} x - 1\right)\left(2\log_{10} x + 5\right) = 0$**
**or $x = 10$ or $2\log_{10} x + 5 = 0$**

**Q.4 Answer:**

$\frac{(2n+2)!}{[(n+1)!]^2}$

**Solution:**

**The coefficient of $(\mathrm{r}+1)^{\mathrm{th}}$ term in the expansion of $(1+\mathrm{x})^{\mathrm{n}+2}$ will be maximum.**
**If $r \leq \frac{(2n+2)+1}{2}$**
$\mathrm{r} \leq (\mathrm{n}+1)+1/2$
$\mathrm{r} = \mathrm{n}+1 =$ **Maximum coefficient** $= {}^{2\mathrm{n}+2}\mathrm{C}_{\mathrm{n}+1} = \frac{(2\mathrm{n}+2)!}{(\mathrm{n}+1)!(\mathrm{n}+1)!} = \frac{(2\mathrm{n}+2)!}{[(\mathrm{n}+1)!]^2}$

**Q.5 Answer:**

10,000

**Solution:**

$101^{100} - 1 = (100+1)^{100} - 1$
$= 100 + 100\mathrm{C}_1 100^{99} + 100\mathrm{C}_2 100^{98} + \ldots + 1 - 1$
$= 100^{100+100}\mathrm{C}_1 100^{99} + 100\mathrm{C}_2 100^{98} + \ldots + {}^{100}\mathrm{C}_{99} 100^1$
$= 100\left(100^{99} + {}^{100}\mathrm{C}_1 100^{98+} \ldots . + {}^{100}\mathrm{C}_{99}\right)$
$= 100\left(100^{99} + 100\mathrm{C}_1 100^{98} + \ldots + {}^{100}\mathrm{C}_{98} 100 + 100\mathrm{C}_{99}\right)$
$= 100\left(100^{99} + {}^{100}\mathrm{C}_1 100^{98} + \ldots . + {}^{100}\mathrm{C}_{98} 100 + 100\right)$
$= 100^2\left(100^{98} + 100\mathrm{C}_1 100^{97} + \ldots + {}^{100}\mathrm{C}_2 + 1\right)$
**$\therefore$ the greatest integer which divides given number $= 100^2 = 10,000$**

**Q.6 Answer:**

41

**Solution:**

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**Here** $T_{r+1} = {}^{10}C_r(\sqrt{2})^{10-r}\left(3^{1/5}\right)^r$, **where** $r = 0, 1, 2, \ldots, 10$

**We observe that in general term** $T_{r+1}$ **powers of 2 and 3 are** $\frac{1}{2}(10-r)$ **and** $\frac{1}{5}r$ **respectively and** $0 \le r \le 10$.

**So both these powers will be integers together only when** $r = 0$ **or 10**

$\therefore$ **Sum of required terms** $= T_1 + T_{11} = {}^{10}C_0(\sqrt{2})^{10} + {}^{10}C_{10}\left(3^{1/5}\right)^{10} = 32 + 9 = 41$

**Q.7 Answer:**

7

**Solution:**

**General term** $= \frac{8!}{\alpha!\beta!\gamma!} \cdot 1^{\alpha} \cdot \left(x^2\right)^{\beta}(-1)^{\gamma}\left(x^3\right)^{\gamma}$

**Where** $\alpha + \beta + \gamma = 8$

**We have** $2\beta + 3\gamma = 10$

**So,** $(\beta, \gamma) \equiv (2, 2)$ **or** $(5, 0)$

**So; coefficient of** $x^{10}$ **is** $\frac{8!}{4!2!2!} + \frac{8!}{3!5!0!} = 476$

**Q.8 Answer:**

210

**Solution:**

$\frac{C_r}{C_{r-1}} = \frac{n-r+1}{r} = \frac{20-r+1}{r} = \frac{21-r}{r}$

$\Rightarrow r \cdot \frac{C_r}{C_{r-1}} = (21-r)$

$\Rightarrow \sum_{r=1}^{20} r \cdot \frac{C_r}{C_{r-1}} = \sum_{r=1}^{20}(21-r)$

$= 21\sum_{r=1}^{20} 1 - \sum_{r=1}^{20} r$

$= 21 \times 20 - \frac{20 \times 21}{2}$

$= 210$

**Q.9 Answer:**

I is a natural number

**Solution:**

$I + f = (4+\sqrt{15})^n$

**Let** $f' = (4-\sqrt{15})^n$. **Then,** $0 < f' < 1$

$I + f = {}^nC_0 4^n + {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 + {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots..$

$f' = {}^nC_0 4^n - {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 - {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots.$

$\therefore \quad I + f + f' = 2\left({}^nC_0 4^n + {}^nC_2 4^{n-2} \times 15 + \ldots.\right) =$ **even integer.**

$\because \quad 0 < f + f' < 2 \Rightarrow f + f' = 1 \Rightarrow 1 - f = f'$

**Thus, I is an odd integer. Now,** $1 - f = f' = (4-\sqrt{15})^n$

$(I + f)(1 - f) = (I + f)f' = 1$

**Q.10 Answer:**

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5

**Solution:**

**Coefficients of $r^{\text{th}}$, $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms ${}^{14}C_{r-1}, {}^{14}C_r$ and ${}^{14}C_{r+1}$**

If these coefficients are in A.P., then $2\left({}^{14}\text{C}_\text{r}\right) = {}^{14}\text{C}_{\text{r}-1}$ and ${}^{14}\text{C}_{\text{r}+1}$

$$\Rightarrow \frac{2(14)!}{\text{r!}(14-\text{r})!} = \frac{(14)!}{(\text{r}-1)!(15-\text{r})!} + \frac{(14)!}{(\text{r}+1)!(13-\text{r})!}$$

$$\Rightarrow \frac{2(14)!}{\text{r!}(14-\text{r})!} = \frac{(14)![(\text{r}+1)\text{r} + (15-\text{r})(14-\text{r})]}{(\text{r}-1)!(15-\text{r})!} \Rightarrow 2(15-\text{r})(\text{r}+1) = 2\text{r}^2 - 28\text{r} + 210$$

$$\Rightarrow \text{r}^2 - 14\text{r} + 45 = 0 \text{ or } (\text{r}-5)(\text{r}-9) = 0 \Rightarrow \text{r} = 5 \text{ or } 9$$

**Q.11 Answer:**

21

**Solution:**

$\frac{{}^{\text{n}}\text{C}_\text{k}}{{}^{\text{n}}\text{C}_{\text{k}+1}} = \frac{1}{2} \Rightarrow \frac{\text{n!}}{\text{k!}(\text{n}-\text{k})!} \frac{(\text{k}+1)!(\text{n}-\text{k}-1)!}{\text{n!}} = \frac{1}{2}$

**or** $\frac{\text{k}+1}{\text{n}-\text{k}} = \frac{1}{2}$

$2k + 2 = n - k$

$\text{n} - 3\text{k} = 2$ **...(1)**

**Similarly** $\frac{{}^{\text{n}}\text{C}_{\text{k}+1}}{{}^{\text{n}}\text{C}_{\text{k}+2}} = \frac{2}{3}$

$\frac{\text{n!}}{(\text{k}+1)!(\text{n}-\text{k}-1)!} \cdot \frac{(\text{k}+2)!(\text{n}-\text{k}-2)!}{\text{n!}} = \frac{2}{3}$

$\frac{\text{k}+2}{\text{n}-\text{k}-1} = \frac{2}{3}$

$3\text{k} + 6 = 2\text{n} - 2\text{k} - 2$

$2n - 5k = 8$ **...(2)**

**From (1) and (2)**

$\text{n} = 14$ **and** $\text{k} = 4$

$\therefore \quad \text{n} + \text{k} = 18$

**Q.12 Answer:**

$-2^{99}$

**Solution:**

$$\text{a}_0 + \text{a}_1\text{x} + \text{a}_2\text{x}^2 + \ldots\ldots + \text{a}_{99}\text{x}^{99} + \text{x}^{100} = 0 \text{ has roots } {}^{99}\text{C}_0, {}^{99}\text{C}_1, {}^{99}\text{C}_2, \ldots\ldots,, {}^{99}\text{C}_{99}$$

$$\Rightarrow \text{a}_0 + \text{a}_1\text{x} + \text{a}_2\text{x}^2 + \ldots\ldots + \text{a}_{99}\text{x}^{99} + \text{x}^{100} = \left(\text{x} - {}^{99}\text{C}_0\right)\left(\text{x} - {}^{99}\text{C}_1\right)\left(\text{x} - {}^{99}\text{C}_2\right)\ldots\ldots\left(\text{x} - {}^{99}\text{C}_{99}\right)$$

**Now, sum of roots is**

$${}^{99}\text{C}_0 + {}^{99}\text{C}_1 + {}^{99}\text{C}_2 + \ldots\ldots + {}^{99}\text{C}_{99} = -\frac{\text{a}_{99}}{\text{coefficient of x}^{100}} \Rightarrow \text{a}_{99} = -2^{99}$$

**Also, sum of product of roots taken two at a time is** $\frac{a_{98}}{\text{coefficient of } x^{100}}$

$$\therefore \quad \sum_{0\le i<j\le 99} \sum_j^{99} C_i {}^{99}C_j = \frac{\left(\sum_{i=0}^{99}\sum_{j=0}^{99} {}^{99}C_i {}^{99}C_j\right)' - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2}$$

$$= \frac{\left(\sum_{i=0}^{99} {}^{99}C_i 2^{99}\right) - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{99}2^{99} - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{198} - {}^{198}C_{99}}{2}$$

$$\left({}^{99}C_0\right)^2 + \left({}^{99}C_1\right)^2 + \ldots. + \left({}^{99}C_{99}\right)^2$$

$$= {}^{99}C_0 + {}^{99}C_1 + {}^{99}C_2 + \ldots\ldots + {}^{99}C_{99}\Big)^2 - 2\sum_{0\le i<j\le 99} {}^{99}C_i {}^{99}C_j = (-a_{99})^2 - 2a_{98} = a_{99}^2 - 2a_{98}$$

**Q.13 Answer:**

$\frac{2^{198} - {}^{198}\text{C}_{99}}{2}$

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**Solution:**

$$a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = 0 \text{ has roots } {}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \ldots\ldots,, {}^{99}C_{99}$$

$$\Rightarrow a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = \left(x - {}^{99}C_0\right)\left(x - {}^{99}C_1\right)\left(x - {}^{99}C_2\right)\ldots\ldots\left(x - {}^{99}C_{99}\right)$$

**Now, sum of roots is**

$${}^{99}C_0 + {}^{99}C_1 + {}^{99}C_2 + \ldots\ldots + {}^{99}C_{99} = -\frac{a_{98}}{\text{coefficient of } x^{100}} \Rightarrow a_{99} = -2^{99}$$

**Also, sum of product of roots taken two at a time is** $\frac{a_{99}}{\text{coefficient of } x^{100}}$

$$\therefore \sum_{0\le i<j\le 99}\sum_{j}^{99} {}^{99}C_i\,{}^{99}C_j = \frac{\left(\sum_{i=0}^{99}\sum_{j=0}^{99} {}^{99}C_i\,{}^{99}C_j\right)' - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2}$$

$$= \frac{\left(\sum_{i=0}^{99} {}^{99}C_i 2^{99}\right) - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{99}2^{99} - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{198} - {}^{198}C_{99}}{2}$$

$$\left({}^{99}C_0\right)^2 + \left({}^{99}C_1\right)^2 + \ldots. + \left({}^{99}C_{99}\right)^2$$

$$= {}^{99}C_0 + {}^{99}C_1 + {}^{99}C_2 + \ldots\ldots + {}^{99}C_{99}\Big)^2 - 2\sum_{0\le i<j\le 99} {}^{99}C_i\,{}^{99}C_j = (-a_{99})^2 - 2a_{98} = a_{99}^2 - 2a_{98}$$

**Q.14 Answer:**

$a_{99}^2 - 2a_{98}$

**Solution:**

$$a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = 0 \text{ has roots } {}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \ldots\ldots,, {}^{99}C_{99}$$

$$\Rightarrow a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{99}x^{99} + x^{100} = \left(x - {}^{99}C_0\right)\left(x - {}^{99}C_1\right)\left(x - {}^{99}C_2\right)\ldots\ldots\left(x - {}^{99}C_{99}\right)$$

**Now, sum of roots is**

$${}^{99}C_0 + {}^{99}C_1 + {}^{99}C_2 + \ldots\ldots + {}^{99}C_{99} = -\frac{a_{99}}{\text{coefficient of } x^{100}} \Rightarrow a_{98} = -2^{99}$$

**Also, sum of product of roots taken two at a time is** $\frac{a_{99}}{\text{coefficient of } x^{100}}$

$$\therefore \sum_{0\le i<j\le 99}\sum_{j}^{99} {}^{99}C_i\,{}^{99}C_j = \frac{\left(\sum_{i=0}^{99}\sum_{j=0}^{99} {}^{99}C_i\,{}^{99}C_j\right)' - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2}$$

$$= \frac{\left(\sum_{i=0}^{99} {}^{99}C_i 2^{99}\right) - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{99}2^{99} - \sum_{i=0}^{99}\left({}^{99}C_i\right)^2}{2} = \frac{2^{198} - {}^{198}C_{99}}{2}$$

$$\left({}^{99}C_0\right)^2 + \left({}^{99}C_1\right)^2 + \ldots. + \left({}^{99}C_{99}\right)^2$$

$$\left(= {}^{99}C_0 + {}^{99}C_1 + {}^{99}C_2 + \ldots\ldots + {}^{99}C_{99}\right)^2 - 2\sum_{0\le i<j\le 99} {}^{99}C_i\,{}^{99}C_j = (-a_{99})^2 - 2a_{98} = a_{99}^2 - 2a_{98}$$

**Q.15 Answer:**

-1

**Solution:**

$2^{\text{nd}}$ **term is** ${}^nC_1x^{n-1}a = 240$ **....(1)**

$3^{\text{rd}}$ **term is** ${}^nC_2x^{n-2}a^2 = 720$ **....(2)**

$4^{\text{th}}$ **term is** ${}^nC_3x^{n-3}a^3 = 1080$ **....(3)**

**Multiplying (1) and (3) and dividing by the square of (2), we get**

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$$\frac{{}^{n}C_1 \times {}^{n}C_3}{({}^{n}C_2)^2} = \frac{240 \times 1080}{(720)^2}$$

$$\Rightarrow \frac{n \times n(n-1)(n-2)(2!)^2}{n^2(n-1)^2 \times 3!} = \frac{1}{2} \Rightarrow 4(n-2) = 3(n-1) \quad (\because n \neq 1)$$

$\Rightarrow \quad n = 5$

Putting $n = 5$, from (1) and (2), we get $5x^4a = 240$ and $10x^3a^2 = 720$

$$\Rightarrow \frac{(5x^4a)^2}{10x^3a^2} = \frac{(240)^2}{720} \text{ or } x^5 = 32 \quad \therefore x = 2$$

$$\therefore \quad a = \frac{240}{5x^4} = \frac{48}{2^4} = 3$$

Hence, $x = 2, a = 3$ and $n = 5$.

$(x-a)^n = (2-3)^5 = -1$

Also, $(2+3)^5 = 2^5 + 5C_1 2^4 \times 3 + {}^{5}C_2 2^3 \times 3^2 + {}^{5}C_3 2^2 \times 3^3 + {}^{5}C_4 2 \times 3^4 + {}^{5}C_5 3^5$

$= 32 + 240 + 720 + 1080 + 810 + 243$

**Hence, least value of the term is 32 .**

**Sum of odd-numbered terms is $32 + 720 + 810 = 1562$**

**Q.16 Answer:**

32

**Solution:**

**$2^{nd}$ term is** ${}^{n}C_1 x^{n-1} a = 240$ **....(1)**

**$3^{rd}$ term is** ${}^{n}C_2 x^{n-2} a^2 = 720$ **....(2)**

**$4^{th}$ term is** ${}^{n}C_3 x^{n-3} a^3 = 1080$ **....(3)**

**Multiplying (1) and (3) and dividing by the square of (2), we get**

$$\frac{{}^{n}C_1 \times {}^{n}C_3}{({}^{n}C_2)^2} = \frac{240 \times 1080}{(720)^2}$$

$$\Rightarrow \frac{n \times n(n-1)(n-2)(2!)^2}{n^2(n-1)^2 \times 3!} = \frac{1}{2} \Rightarrow 4(n-2) = 3(n-1) \quad (\because n \neq 1)$$

$\Rightarrow \quad n = 5$

Putting $n = 5$, from (1) and (2), we get $5x^4a = 240$ and $10x^3a^2 = 720$

$$\Rightarrow \frac{(5x^4a)^2}{10x^3a^2} = \frac{(240)^2}{720} \text{ or } x^5 = 32 \quad \therefore x = 2$$

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$= 32 + 240 + 720 + 1080 + 810 + 243$

**Hence, least value of the term is 32 .**

**Sum of odd-numbered terms is $32 + 720 + 810 = 1562$**

**Q.17 Answer:**

1562

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**Solution:**

$2^{\text{nd}}$ **term is** $^{n}C_1 x^{n-1} a = 240$ **....(1)**

$3^{\text{rd}}$ **term is** $^{n}C_2 x^{n-2} a^2 = 720$ **....(2)**

$4^{\text{th}}$ **term is** $^{n}C_3 x^{n-3} a^3 = 1080$ **....(3)**

**Multiplying (1) and (3) and dividing by the square of (2), we get**

$$\frac{{}^{\mathrm{n}}\mathrm{C}_1 \times {}^{\mathrm{n}}\mathrm{C}_3}{\left({}^{\mathrm{n}}\mathrm{C}_2\right)^2} = \frac{240 \times 1080}{(720)^2}$$

$$\Rightarrow \frac{\mathrm{n} \times \mathrm{n}(\mathrm{n}-1)(\mathrm{n}-2)(2!)^2}{\mathrm{n}^2(\mathrm{n}-1)^2 \times 3!} = \frac{1}{2} \Rightarrow 4(\mathrm{n}-2) = 3(\mathrm{n}-1) \quad (\because \mathrm{n} \neq 1)$$

$$\Rightarrow \quad \mathrm{n} = 5$$

$$\text{Putting } \mathrm{n} = 5, \text{ from } (1) \text{ and } (2), \text{ we get } 5\mathrm{x}^4\mathrm{a} = 240 \text{ and } 10\mathrm{x}^3\mathrm{a}^2 = 720$$

$$\Rightarrow \frac{\left(5\mathrm{x}^4\mathrm{a}\right)^2}{10\mathrm{x}^3\mathrm{a}^2} = \frac{(240)^2}{720} \text{ or } \mathrm{x}^5 = 32 \quad \therefore \mathrm{x} = 2$$

$$\therefore \quad \mathrm{a} = \frac{240}{5\mathrm{x}^4} = \frac{48}{2^4} = 3$$

$$\text{Hence, } \mathrm{x} = 2, \mathrm{a} = 3 \text{ and } \mathrm{n} = 5.$$

$$(\mathrm{x}-\mathrm{a})^{\mathrm{n}} = (2-3)^5 = -1$$

$$\text{Also, } (2+3)^5 = 2^5 + 5\mathrm{C}_1 2^4 \times 3 + {}^{5}\mathrm{C}_2 2^3 \times 3^2 + {}^{5}\mathrm{C}_3 2^2 \times 3^3 + {}^{5}\mathrm{C}_4 2 \times 3^4 + {}^{5}\mathrm{C}_5 3^5$$

$$= 32 + 240 + 720 + 1080 + 810 + 243$$

**Hence, least value of the term is 32 .**

**Sum of odd-numbered terms is** $32 + 720 + 810 = 1562$

**Q.18 Answer:**

(A) - q, (B) - s, (C) - p, (D) - r

**Solution:**

$(\mathrm{A}) - \mathrm{q}, (\mathrm{B}) - \mathrm{s}, (\mathrm{C}) - \mathrm{p}, (\mathrm{D}) - \mathbf{r}$

**We know that**

$$^{n}C_0^2 + {}^{n}C_1^2 + \ldots.. + {}^{n}C_n^2 = {}^{2n}C_n$$

**and** $^{\mathrm{n}}C_0^2 + {}^{\mathrm{n}}C_1^2 + \ldots.. + {}^{\mathrm{n}}C_{\mathrm{n}}^2 = \begin{cases} 0, & \text{if n is odd} \\ {}^{\mathrm{n}}C_{\mathrm{n}/2}(-1)^{\mathrm{n}}, & \text{if n is even} \end{cases}$

**From this,**

$^{31}C_0^2 - {}^{31}C_1^2 + {}^{31}C_2^2 - \ldots.. - {}^{31}C_{31}^2 = 0\ {}^{32}\mathrm{C}_0^2 - {}^{32}\mathrm{C}_1^2 + {}^{32}\mathrm{C}_2^2 - \ldots. + {}^{32}\mathrm{C}_{32}^2 = {}^{32}\mathrm{C}_{16}\ {}^{32}C_0^2 + {}^{32}C_1^2 + {}^{32}C_2^2 - \ldots.. + {}^{32}C_{32}^2 = {}^{64}C_{32}$

**Also,** $\frac{1}{32}\left(1 \times {}^{32}C_1^2 + 2 \times {}^{32}C_2^2 + \ldots. + 32 \times {}^{32}C_{32}^2\right) = \frac{1}{32}\sum_{r=1}^{32} r\left({}^{32}C_r\right)^2$

$= \frac{1}{32}\sum_{r=1}^{32} r\,{}^{32}C_r\,{}^{32}C_{32-r} = \frac{1}{32}\sum_{r=1}^{32} 32\,{}^{31}C_{r-1}\,{}^{32}C_{32-r} = {}^{63}C_{31} = {}^{63}C_{32}$

**Q.19 Answer:**

—

**Solution:**

**Coefficient of** $x^{-5}$ **in** $\left(2x^2 - \frac{1}{x}\right)^8 = a$**, (say)**

**Coefficient of** $x$ **in** $\left(2x^2 - \frac{1}{x}\right)^8 = b$**, (say)**

**Term independent of** $x$ **in** $\left(2x^2 - \frac{1}{x}\right)^s = c$**, (say)**

$\mathrm{T}_{r+1}$ **of** $\left(2x^2 - \frac{1}{x}\right)^8 = {}^{8}\mathrm{C}_r\left(2x^2\right)^{8-r}\left(-\frac{1}{x}\right)^r$

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$= {}^{8}\mathrm{C}_{r}(-1)^{r}(2)^{8-r}(x)^{16-3r}$

**Coefficient of $x^{-5}$,**

$16-3r=-5 \Rightarrow r=7$

$\Rightarrow a = {}^{8}\mathrm{C}_{7}(-1)^{7}(2)^{8-7} = -16$

**Coefficient of $x$,**

$16-3r=1 \Rightarrow r=5$

$\Rightarrow b = {}^{8}\mathrm{C}_{5}(-1)^{5}(2)^{8-5} = -448$

**There is no term which is independent of $x$ in $\left(2x^{2}-\frac{1}{x}\right)^{8}$**

$\Rightarrow$ **Term independent of $x = c-b+3a$**

$= 0+448-48$

$= 400$

**Q.20 Answer:**

—

**Solution:**

$$\left(x+\sqrt{x^{3}-1}\right)^{5}+\left(x-\sqrt{x^{3}-1}\right)^{5}=2\left[x^{5}+{}^{5}C_{2}\cdot x^{3}\left(x^{3}-1\right)+{}^{5}C_{4}x\left(x^{3}-1\right)^{2}\right]$$

$$=2\left[x^{5}+10x^{3}\left(x^{3}-1\right)+5x\left(x^{6}-2x^{3}+1\right)\right]$$

$$=10x^{7}+20x^{6}+2x^{5}-20x^{4}-20x^{3}+10x$$

$\therefore$ Polynomial has order of 7.

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